

Prashanta Kumar Roy

PG (I Year I Semester)
M.Tech. Chemical Engineering
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Registration No: 26561029/2027

Indian Institute of Technology Roorkee

VERIFIED

Area of Interest

Upstream Operations, Downstream Refining and Distillation, Simulation and Modeling, Process Engineering and Design

Education

Year	Degree/Examination	Institution/Board	CGPA/ Percentage
2024	Graduate (UG)	BINESWAR BRAHMA ENGINEERING COLLEGE, KOKRAJHAR, ASSAM	8.320
2020	Intermediate (Class XII)	DELHI PUBLIC SCHOOL, DHALIGAON, ASSAM	88.80 %
2018	Matriculate (Class X)	DELHI PUBLIC SCHOOL, DHALIGAON, ASSAM	89.80 %

Internships

Industrial Training Internship | Indian Oil Corporation Limited, Bongaigaon Refinery, Bongaigaon, Assam
July 2023 - August 2023

- Observed operations personnel performing tasks to gain an understanding of their duties.
- Acquired knowledge about crude oil and its products and by-products along with various process equipment used in the refinery.

Mathworks MATLAB | Internshala Trainings October 2021 - November 2021

- Introduction, Foundations, Mathematics, Functions, Plotting, Programming and Case Studies.
- Completed a final project module assigned securing 90% marks.

Social Internship | Naba Nirman Trust July 2021

- Helped bring awareness about cleanliness of streets.

Projects

Extraction of Silica Nanoparticles from Fly Ash | Bineswar Brahma Engineering College, Kokrajhar, Assam
July 2023 - June 2024

- Conducted a project focused on the extraction of silica nanoparticles from coal fly ash, a byproduct from NTPC, Salakati. The process involved water washing, drying, and chemical treatment using a NaOH solution. Key steps included heating in a magnetic stirrer with a condenser, separation of supernatants, titration with HCl, and filtration.
- Final silica nanoparticle samples were characterized through FTIR (Fourier Transform Infrared Spectroscopy) and SEM (Scanning Electron Microscopy) to analyze their structural and compositional properties. This project demonstrated proficiency in laboratory techniques, chemical handling, and nanoparticle characterization methods.

Skills

Computer languages	C, C++, HTML, MATLAB, Python
Software Packages	Mathworks MATLAB, Fusion 360, Blender, DaVinci Resolve, Canva
Additional Courses	Foundations of Business Intelligence by Google
Languages Known	English, Hindi, Assamese

Positions of Responsibility & Extra Curriculars

Intern | North Eastern Hill University August 2022

- Synthesis of different nanoparticles for anti-cancer drug delivery and their characterization techniques

Curriculum Writer | Team Everest NGO September 2021 - October 2021

- Curriculum Development and Instructional Writing

Trainee | Brand Sapien, Qoptars and IICBBEC September 2021

- 3D Printing and Designing on Fusion 360 Software

Participant | TECHNICHE, IIT Guwahati October 2022 - November 2022

- AutoCAD/Catia/Ansys